Clustering Algorithms (k-means and DBSCAN)

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Introduction

- Machine Learning Techniques – 3 Types
  - Supervised
    - **Regression**: predict continuous values.
      - Linear regression
    - **Classification**: predict categories
      - Logistic regression, Decision tree, *k*-Nearest Neighbor
  - Unsupervised
    - Clustering problem: hierarchical clustering, *k*-means clustering
    - Dimensionality reduction problem
  - Reinforcement Learning
    - Reinforcement Learning is a type of machine learning where **an agent (for example, the algorithm)** learns to make decisions by receiving **rewards or penalties** from its environment.

Introduction

- A **common thread in all machine learning techniques** is that they **learn patterns or make decisions by being trained on data**.
  - **Supervised learning:** Trained on labeled data (input → correct output).
  - **Unsupervised learning:** Trained on unlabeled data to find patterns or structure.
  - **Reinforcement learning:** Learns by trial-and-error, receiving feedback (rewards/penalties) from interactions with the environment.

- So, whether it's labeled examples, raw data, or environment feedback, **data or experience drives the learning**.

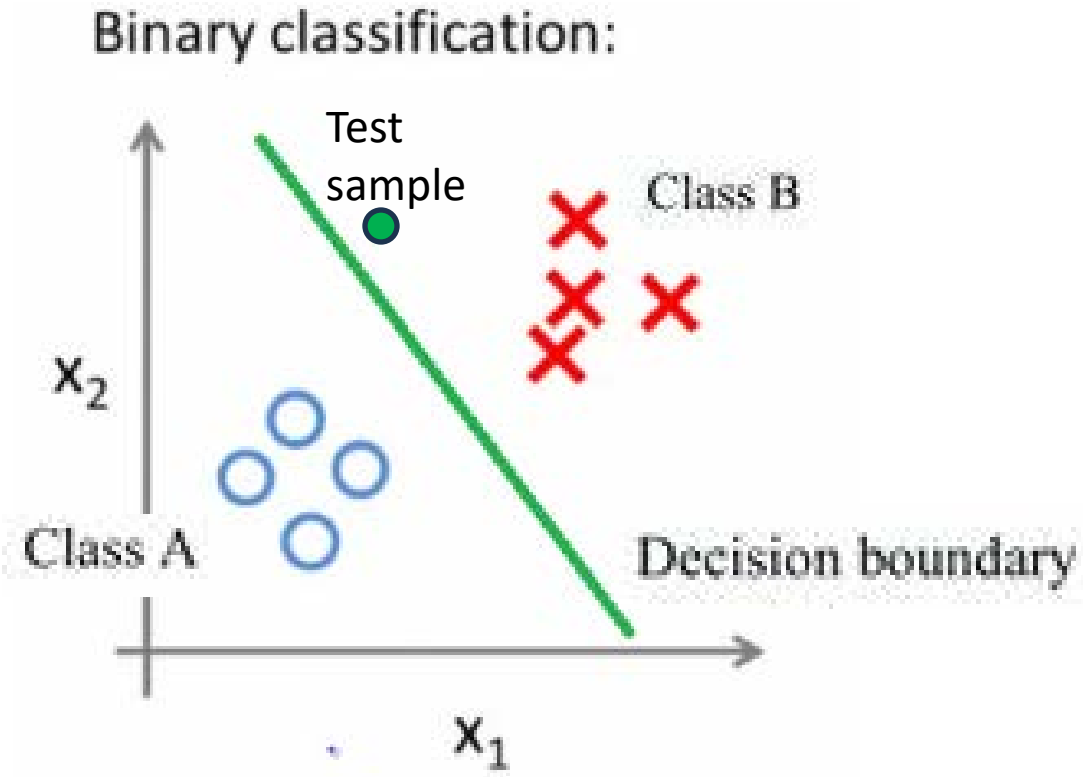
Supervised Techniques

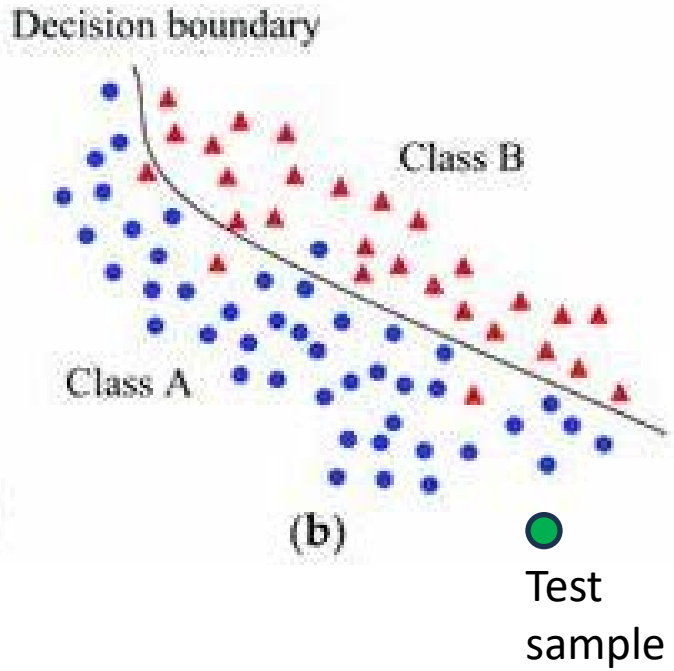
- Nature of Training data
  - **Labeled data**

- Labeled Data:
  - It refers to the data that has been tagged or annotated with **meaningful information (labels)** that describe what each piece of data represents.
    - input → correct output

Supervised Techniques

- Classification





Labeled Data - Examples

- In image recognition (cat *vs.* dog):
  - A photo of a cat 🐱 labeled as **"cat"**.
  - A photo of a dog 🐶 labeled as **"dog"**.

- In sentiment analysis:
  - Text: *"The movie was great!"* → Label: **Positive**.
  - Text: *"The food was terrible."* → Label: **Negative**.

- In medical diagnosis:
  - X-ray image → Label: **"Pneumonia"** or **"Healthy"**.

Labeled database: Example 1 - Play Tennis dataset

Outlook	Temperature	Humidity	Windy	Play
sunny	hot	high	false	no
sunny	hot	high	true	no
overcast	hot	high	false	yes
rainy	mild	high	false	yes
rainy	cool	normal	false	yes
rainy	cool	normal	true	no
overcast	cool	normal	true	yes
sunny	mild	high	false	no
sunny	cool	normal	false	yes
rainy	mild	normal	false	yes
sunny	mild	normal	true	yes
overcast	mild	high	true	yes
overcast	hot	normal	false	yes
rainy	mild	high	true	no

Example: To play tennis or not

Inputs (Features)
→ **First four columns**

Output (Label)
→Last column **Play**

Labeled database: Example 2 – Loan dataset

Applicant Age	Annual Income	Credit Score	Existing Debts	Loan Approved (Label)
28	₹6,00,000	720	No	Yes
45	₹4,50,000	680	Yes	No
32	₹8,00,000	750	No	Yes
50	₹3,00,000	640	Yes	No
38	₹5,50,000	700	No	Yes
26	₹4,00,000	690	No	No

Example: Predicting if a loan application will be approved

Inputs (Features):

→ Age, Income, Credit Score, Debts — measurable and available in real banking systems.

Output (Label):

→ Whether the bank approved the loan or not.

Labeled database: Example 3 - Diabetes dataset

Age	BMI	Blood Pressure (mmHg)	Glucose Level (mg/dL)	Diabetes (Label)
45	28.5	130	180	Yes
33	24.0	120	95	No
50	31.2	140	210	Yes
29	22.1	118	88	No
60	34.8	150	170	Yes
42	26.3	125	102	No

Example: Predicting whether a patient has diabetes.

Inputs (Features):
→ Age, BMI, Blood Pressure, Glucose Level.

Output (Label):
→ Diabetes Test Outcomes: “Yes” or “No” for diabetes diagnosis (based on confirmed medical tests).

Labeled database: Example 4 - Fruit dataset

Color	Weight (grams)	Size (cm)	Sweetness (1-10)	Fruit Type (Label)
Red	180	8	8	Apple
Yellow	120	6	9	Banana
Green	150	7	6	Apple
Orange	160	7	8	Orange
Yellow	100	5	10	Banana
Orange	170	8	7	Orange
Red	200	9	9	Apple

Classification of different fruits

Inputs (Features):
→ Color, Weight, Size, Sweetness level.

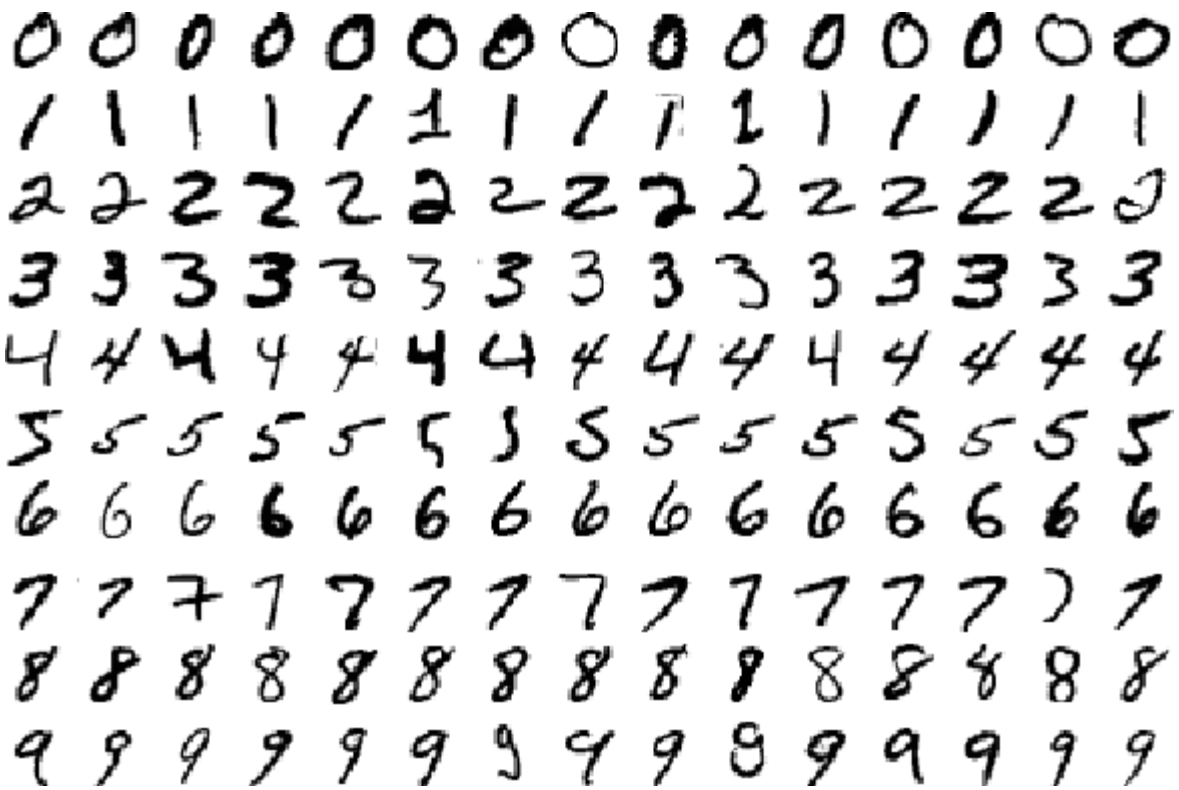
Output (Label):
→ Type of fruit (Apple, Banana, Orange).

Image-based Labeled Public Datasets

Dataset Name	Type	Description	Example Labels
MNIST	Supervised	70,000 grayscale images of handwritten digits (28×28 pixels)	0–9
CIFAR-10	Supervised	60,000 color images (32×32 pixels) of 10 object categories	Airplane, Dog, Truck…
ImageNet	Supervised	14M+ high-resolution images, 20,000+ categories	Cat, Car, Chair…
COCO	Supervised	330,000 images with object detection & segmentation	Person, Bicycle, Pizza…
LFW	Supervised	13,000 face images	Person's name
ChestX-ray14	Supervised	100,000+ chest X-ray images for medical diagnosis	Pneumonia, Healthy…

MNIST Dataset

- Link: http://yann.lecun.com/exdb/mnist/



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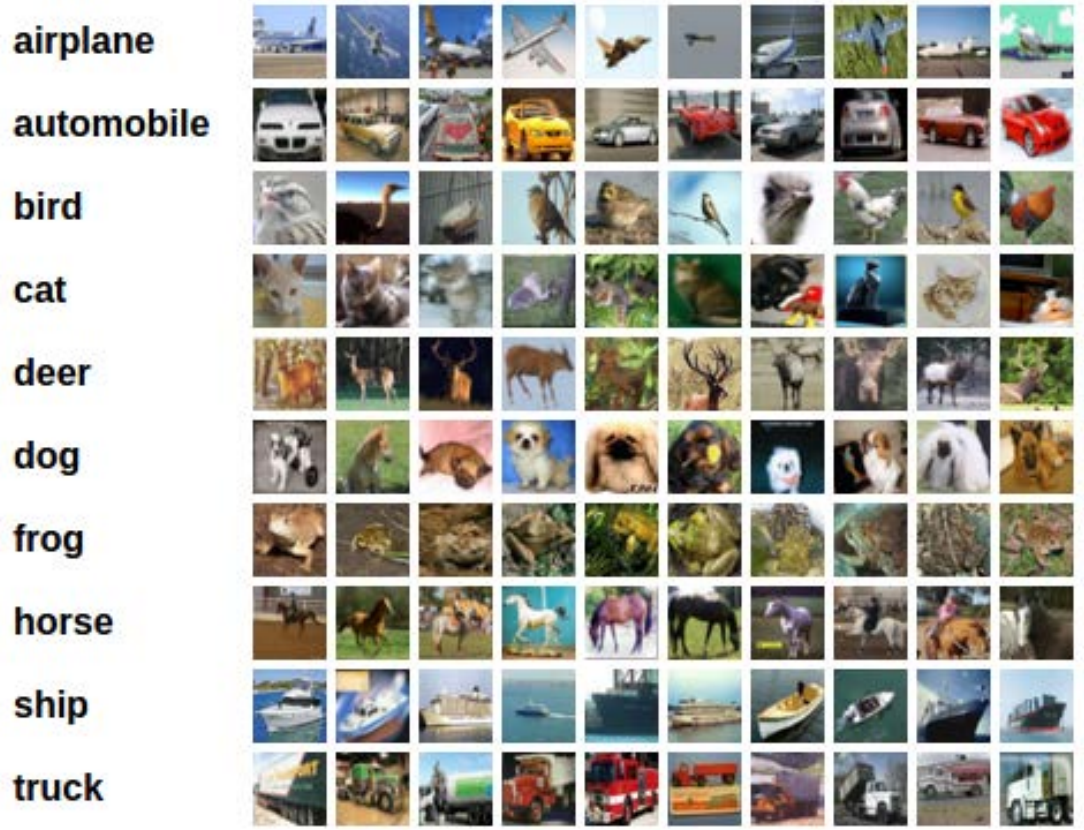
→ Training samples: 60,000
→ Test samples: 10,000

→ Each image is represented by 28x28 pixels,
→ Each containing a value 0 - 255 with its grayscale value.

CIFAR10 Dataset

- It has 10 classes:
  - 'airplane', 'automobile', 'bird', 'cat', 'deer', 'dog', 'frog', 'horse', 'ship', 'truck'.

- The images in CIFAR-10 are of size 3x32x32
  - 3-channel color images of 32x32 pixels in size.



Unsupervised Techniques

- Nature of Training data
  - **No target labels** are provided – the algorithm only has input features.
  - The algorithm tries to **discover hidden patterns** in the data by exploring the similarities in features of different input samples.

Example: Fruit Dataset (Unlabeled)



Discover hidden patterns in the data by exploring the similarities in features of different input samples.

Example: Customer Shopping Data (Unlabeled)

Customer ID	Age	Annual Income (₹)	Spending Score
1	22	25,000	39
2	35	60,000	81
3	26	45,000	6
4	29	30,000	77
5	45	80,000	40
6	34	25,000	5

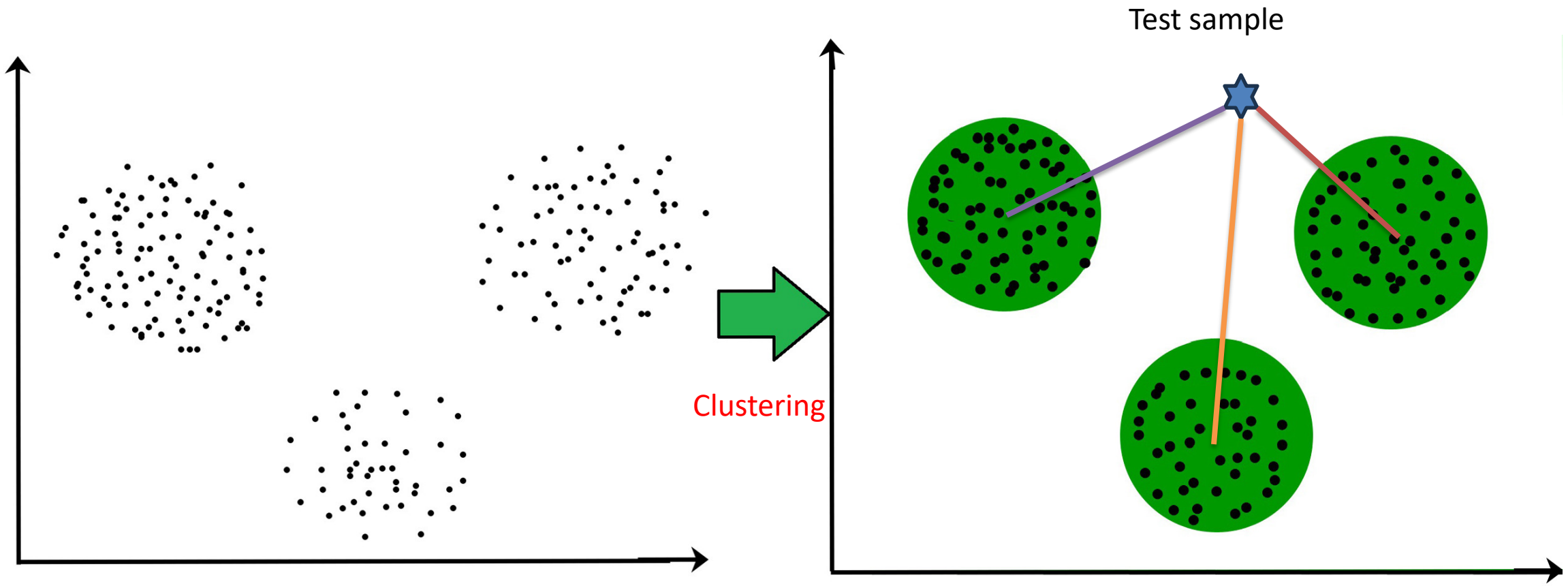
Problem:

→ To group customers with similar spending patterns.

Why is it unsupervised?

- There is **no "target" column** (like "Will Buy Again: Yes/No").
- The model can use clustering (e.g., **K-Means**) to group customers with similar spending patterns.

Clustering Overview



Applications

→ Get insights of the data by analyzing each cluster

→ A new test data instance can be assigned to the most appropriate cluster based on distance or density criteria

Applications of Clustering

- What is the use of grouping the customers with similar spending patterns.

- **1. Targeted Marketing**
  - Instead of sending the same offers to everyone, the company can send **customized promotions** to each group.
  - **Example:**
    - High spenders → Premium product recommendations
    - Low spenders → Discount coupons to encourage more purchases

Applications of Clustering

- **2. Product Recommendations**
  - Suggest products that similar customers bought.
  - **Example:**
    - "Customers like you also purchased…" on Amazon.

- **3. Improve Customer Retention**
  - Identify groups at risk of leaving and give them **special incentives**.
  - **Example:**
    - People with low spending score but high income might need better engagement.

Applications of Clustering

- **4. Strategic Decision-Making**
  - Helps decide **where to open new stores, which products to stock more**, or **which regions to focus on**.

- **5. Personalized Customer Experience**
  - Make customers feel valued by understanding their **preferences**.

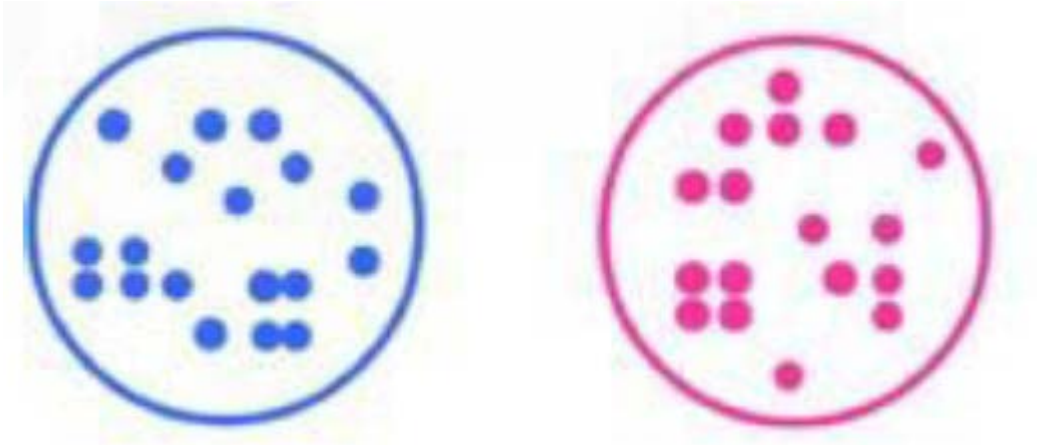
What is Clustering?

- **Clustering** is an **unsupervised machine learning technique** used to group similar items together based on their features, **without using predefined labels**.
  - The idea is that items in the **same cluster** are more similar to each other, while items in **different clusters** are less similar.

- **Example: Shopping Mall Customers**
  - A mall collects data about customers' **age** and **annual income**.
  - A clustering algorithm might group them into:
    - **Cluster 1:** Young with low income (students, interns)
    - **Cluster 2:** Middle-aged with high income (professionals)
    - **Cluster 3:** Elderly with medium income (retirees)

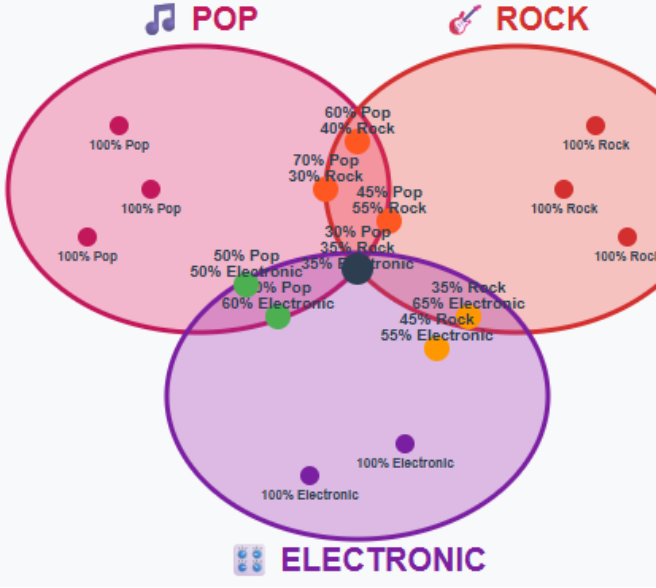
Different Clustering Techniques

- **Exclusive Clustering (Hard Clustering)**
  - Each data point belongs to **exactly one cluster**.
  - **Key Idea:** No overlaps — once assigned, a point can't be part of another cluster.
  - **Example Algorithm:** K-Means, K-Medoids.
  - **Example Use:**
    - Dividing students into groups based on grades (A-group, B-group, etc.) where each student fits only one group.



Different Clustering Techniques

- **Overlapping Clustering (Soft Clustering / Fuzzy Clustering)**
  - A data point can belong to **multiple clusters** with varying degrees of membership (probabilities or weights).
  - **Key Idea:** Real-world objects can share traits with more than one group.
  - **Example Algorithm:** Fuzzy C-Means, Gaussian Mixture Models (GMM).
  - **Example Use:**
    - Music classification → A song can be both "Pop" (60% membership) and "Rock" (40% membership).



Different Clustering Techniques

- **Hierarchical Clustering**
  - Creates a **tree-like structure** (dendrogram) of clusters by merging or splitting them step-by-step.
  - **Types:**
    - **Agglomerative (Bottom-Up):** Start with each point as a cluster → merge.
    - **Divisive (Top-Down):** Start with one cluster → split.

Quick Comparison Table

Feature	Exclusive Clustering	Overlapping Clustering	Hierarchical Clustering
Membership	One cluster only	Multiple clusters	Multiple levels
Output	Flat groups	Membership weights	Dendrogram
Example Algorithm	K-Means	Fuzzy C-Means, GMM	Agglomerative, Divisive
Best For	Clear separation	Shared characteristics	Data with hierarchy

K-Means Clustering

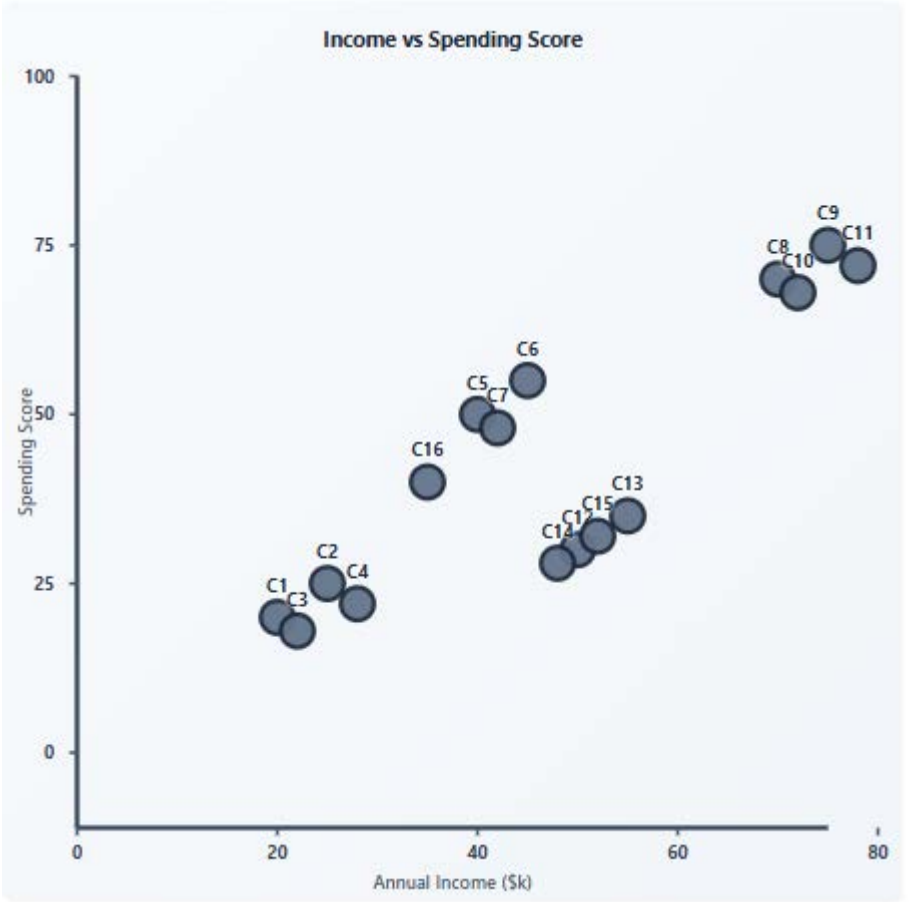
- K-means is an unsupervised machine learning algorithm that groups similar data points together into K clusters.

- Each cluster has a center (centroid), and data points are assigned to the nearest centroid.



K-Means Clustering

- Partition-based clustering algorithm
- Goal: Minimize intra-cluster variance
- Steps: Assign points → Update centroids → Repeat



K-Means Clustering

- Example: Customer data with two features:
  - annual income (in thousands) and
  - spending score (1-100).

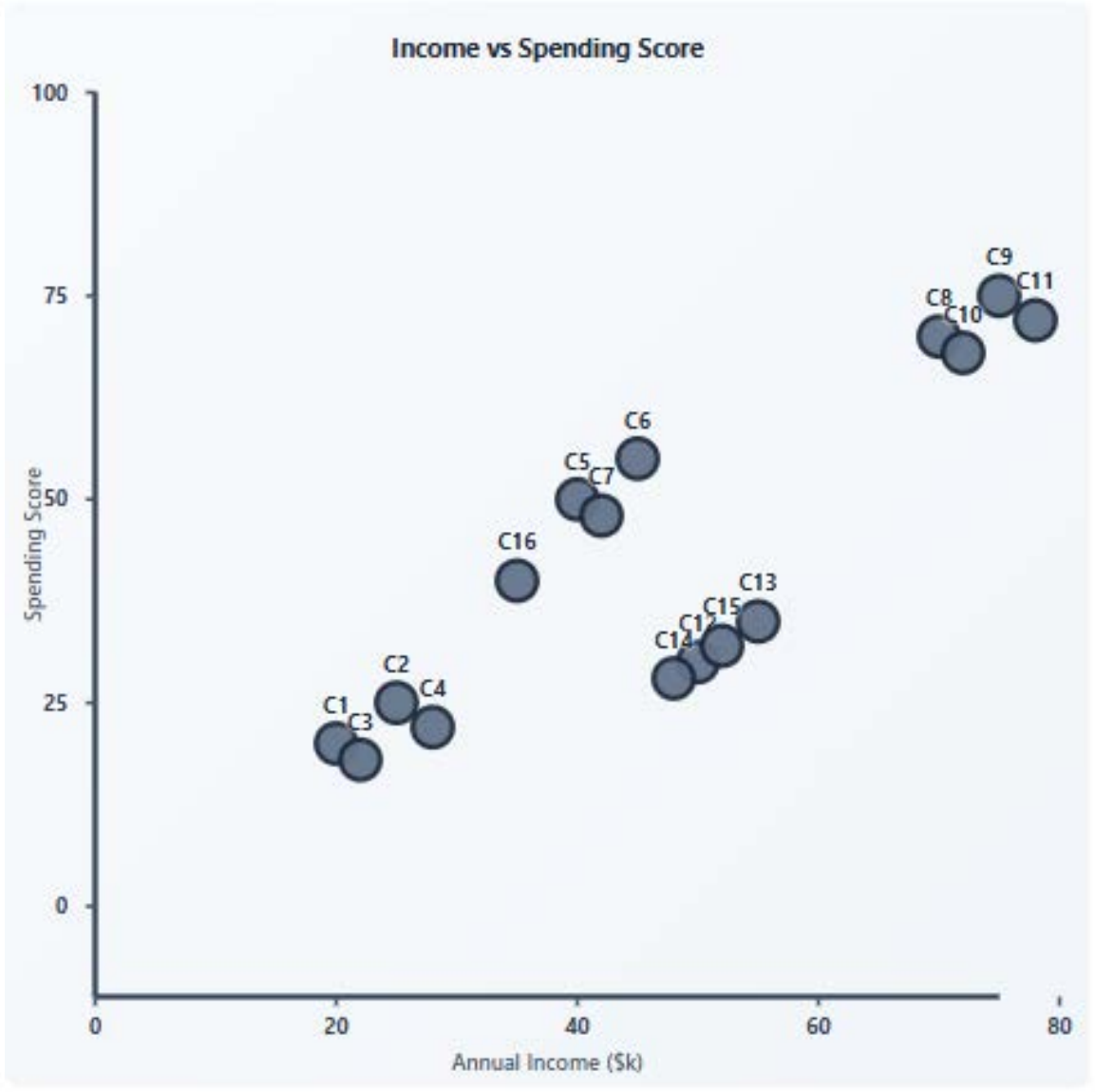
- In this example, we want to get two clusters of similar customers (income and spending pattern)

K-Means Clustering

Customer	Income ($k)	Spending Score
C1	15	39
C2	16	81
C3	17	6
C4	18	77
C5	19	40
C6	70	76
C7	71	94
C8	72	35
C9	14	50
C10	20	85
C11	16	12
C12	15	70
C13	68	88
C14	73	45
C15	69	65
C16	74	80



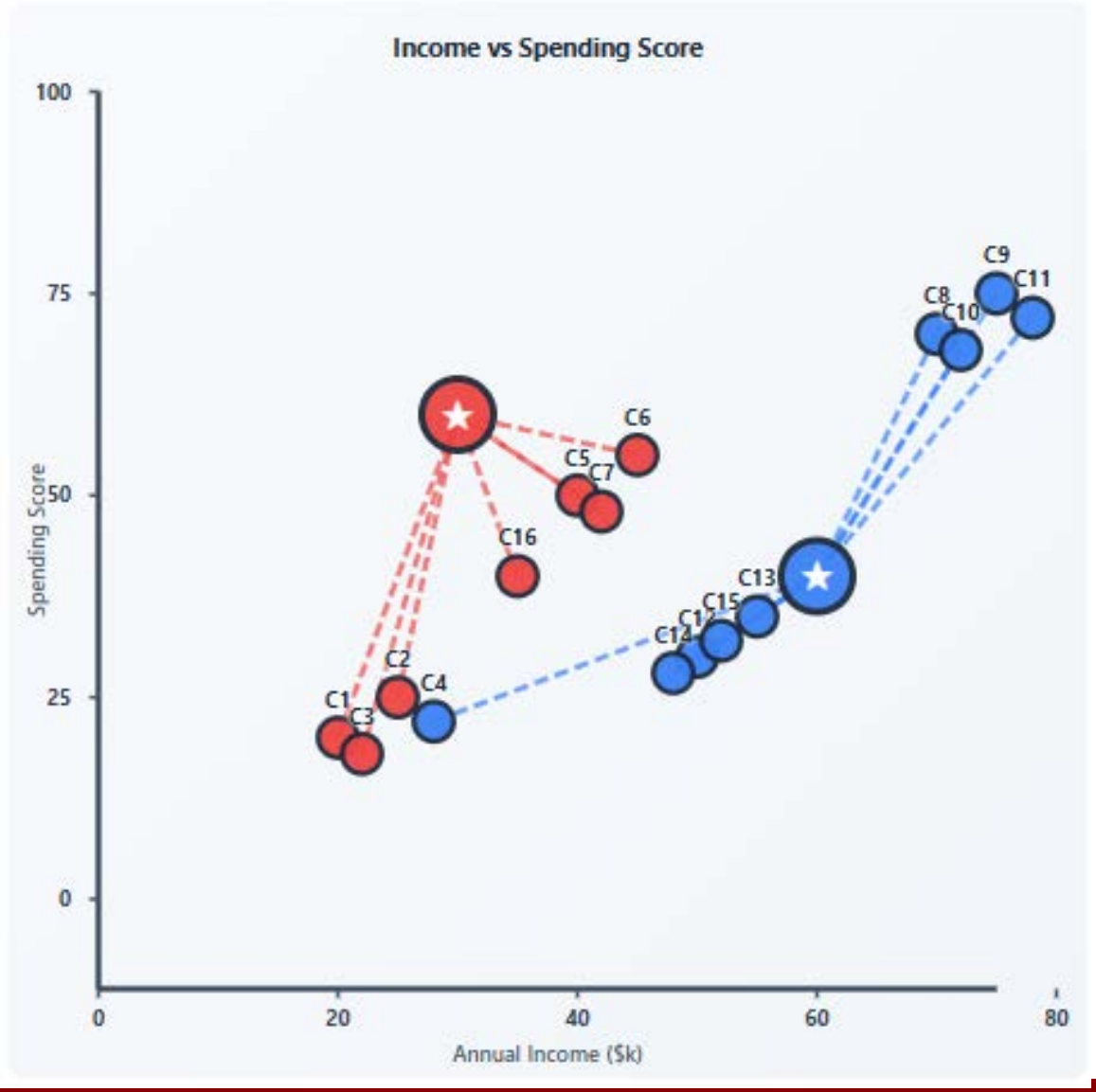
K-Means Clustering – Step 0



K-Means Clustering – Step 1



K-Means Clustering – Step 1



K-Means Clustering – Step 1



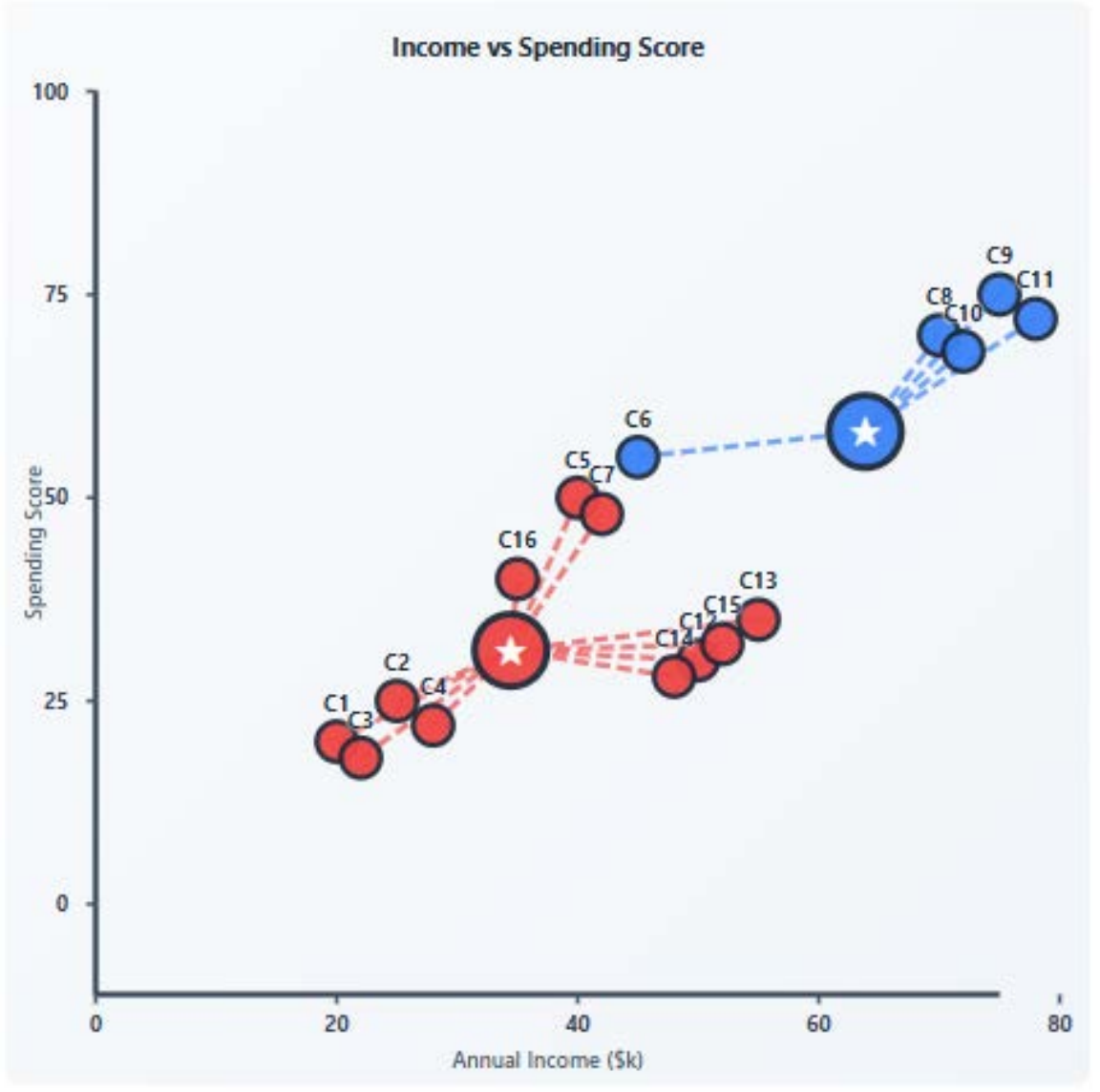
K-Means Clustering – Step 2



K-Means Clustering – Step 2



K-Means Clustering – Step 3



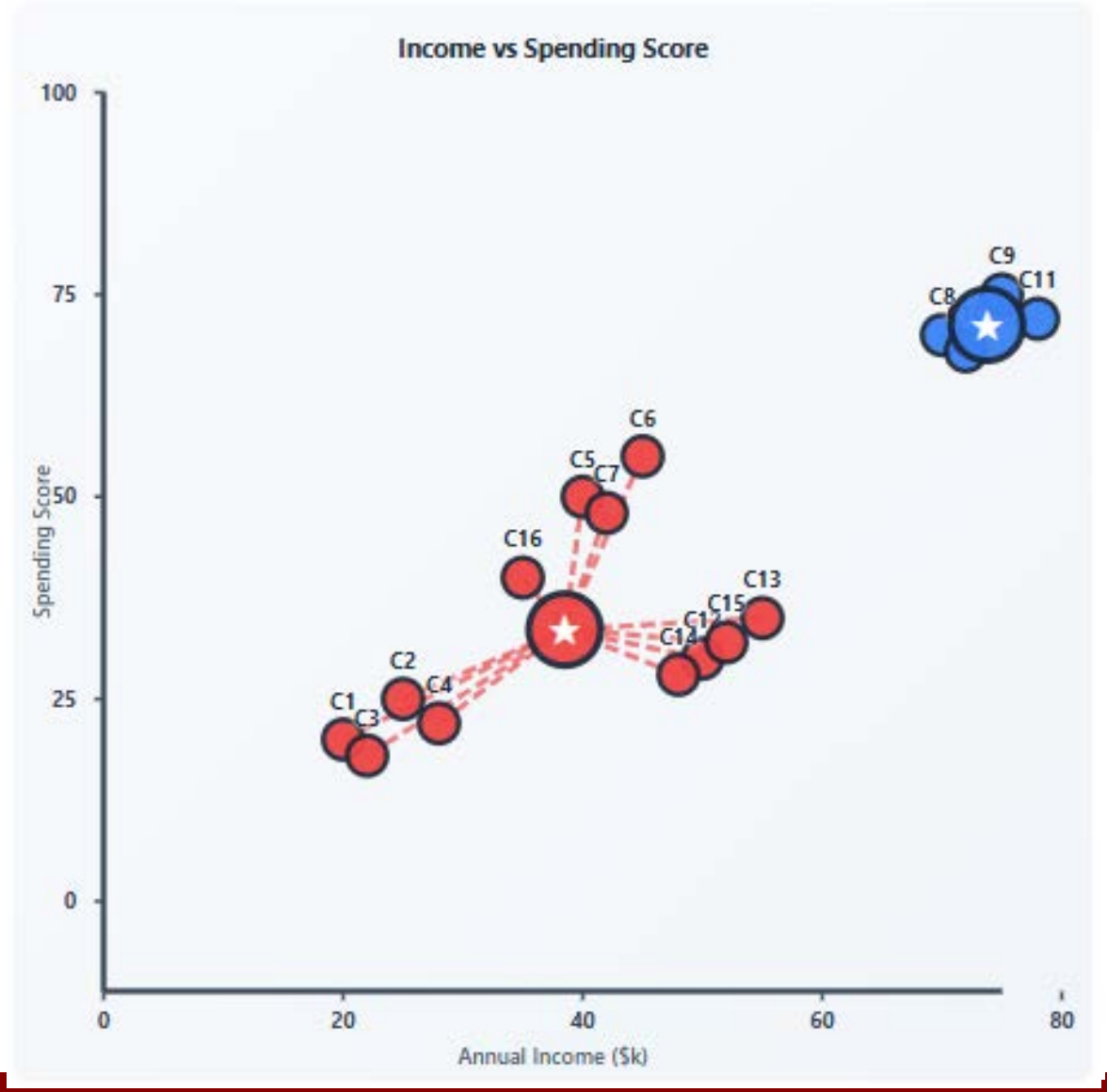
K-Means Clustering – Step 3



K-Means Clustering – Step 4



K-Means Clustering – Step 4



Introduction to K-Means

- **K-Means Workflow**
  - Initialize K cluster centers
  - Assign points to nearest cluster
  - Update cluster centers (mean of points)
  - Repeat until convergence

Batch K-Means Algorithm

Algorithm Steps (Input + Steps + Output):

1. **Input:** Dataset $X = \{x_1, x_2, \ldots, x_n\}$, number of clusters K
2. Initialize centroids $\mu_1, \ldots, \mu_K$ randomly
3. Repeat until convergence:
   - Assign each point x_i to nearest centroid:

$$C_i = \arg\min_j ||x_i - \mu_j||^2$$

   - Update centroid:

$$\mu_j = \frac{1}{|C_j|} \sum_{x \in C_j} x$$

4. **Output:** Final cluster centroids $\mu_1, \ldots, \mu_K$ and cluster assignments C_i

Batch K-Means Algorithm

Algorithm:

1. **Initialize** k cluster centers randomly from the dataset $X = \{x_1, x_2, \ldots, x_n\}$.
2. **Repeat until convergence (or maximum iterations):**

   a. **Assignment Step:** For each point $x_i \in X$, find the nearest cluster center c_j and assign x_i to cluster j.

   b. **Update Step:** For each cluster j, update the cluster center:

$$c_j = \frac{1}{|S_j|} \sum_{x \in S_j} x$$

   where S_j is the set of all points assigned to cluster j.
3. **Stop** when cluster centers do not change significantly (or assignments remain stable).

Advantages and Limitations of Batch K-Means

- Advantages
  - Simple and widely used
  - Deterministic for fixed data
  - Easy to implement

- Limitations
  - Slow for large datasets
  - Requires all data in memory
  - Sensitive to outliers and initial centroids

Online K-Means Algorithm

- Unlike the standard batch K-Means (which updates cluster centroids after all points are assigned in one iteration), the **online (or incremental) K-Means** updates the centroid **immediately after each data point is processed**.

Online K-Means Algorithm

Algorithm:

1. **Initialize** k cluster centers randomly from the dataset $X = \{x_1, x_2, \ldots, x_n\}$.
2. **For each point** $x_i \in X$:

   a. Find the nearest cluster center c_j.

   b. Assign x_i to cluster j.

   c. Update the cluster center:

$$c_j = \frac{1}{|S_j|} \sum_{x \in S_j} x$$

   where S_j is the set of all points assigned so far to cluster j.
3. **Repeat** until all points are processed (or until convergence in the streaming case).

Online K-Means Algorithm

Online k-means Algorithm (update after each point)

Input:

- Dataset $X = \{x_1, x_2, \ldots, x_n\}$
- Number of clusters k

Steps:

1. **Initialize** k cluster centers randomly (choose from the data).
2. For each point $x_i \in X$:

   a. Find the nearest cluster center c_j.

   b. Assign x_i to cluster j.

   c. **Update cluster center** c_j:

$$c_j = \frac{1}{|S_j|} \sum_{x \in S_j} x$$

   where S_j is the set of all points assigned so far to cluster j.
3. Repeat until all points are processed (or until convergence in streaming case).

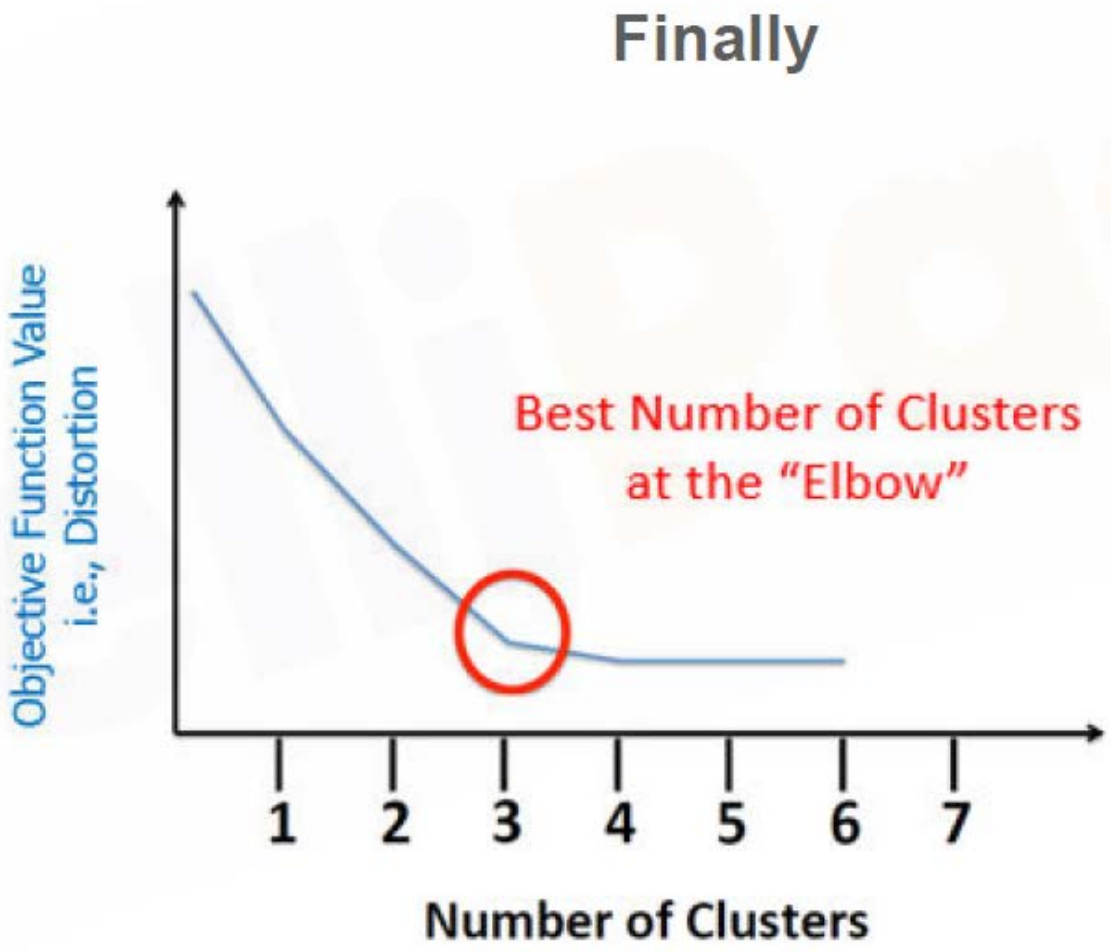
Key Difference from Batch K-Means

- **Batch K-Means:**
  - Updates centroids after processing **all points** in an iteration.
- **Online K-Means:**
  - Updates centroids **immediately** after processing **each point**.

Comparison of Batch vs Online K-Means

Feature	Batch K-Means	Online K-Means
Updates	After all points	After each point
Memory	High	Low
Speed	Slower	Faster for streaming
Use Case	Static data	Streaming/large data

Choosing K - Elbow Method



Applications of K-Means

- Customer segmentation
- Image compression
- Anomaly detection
- Market basket analysis

DBSCAN

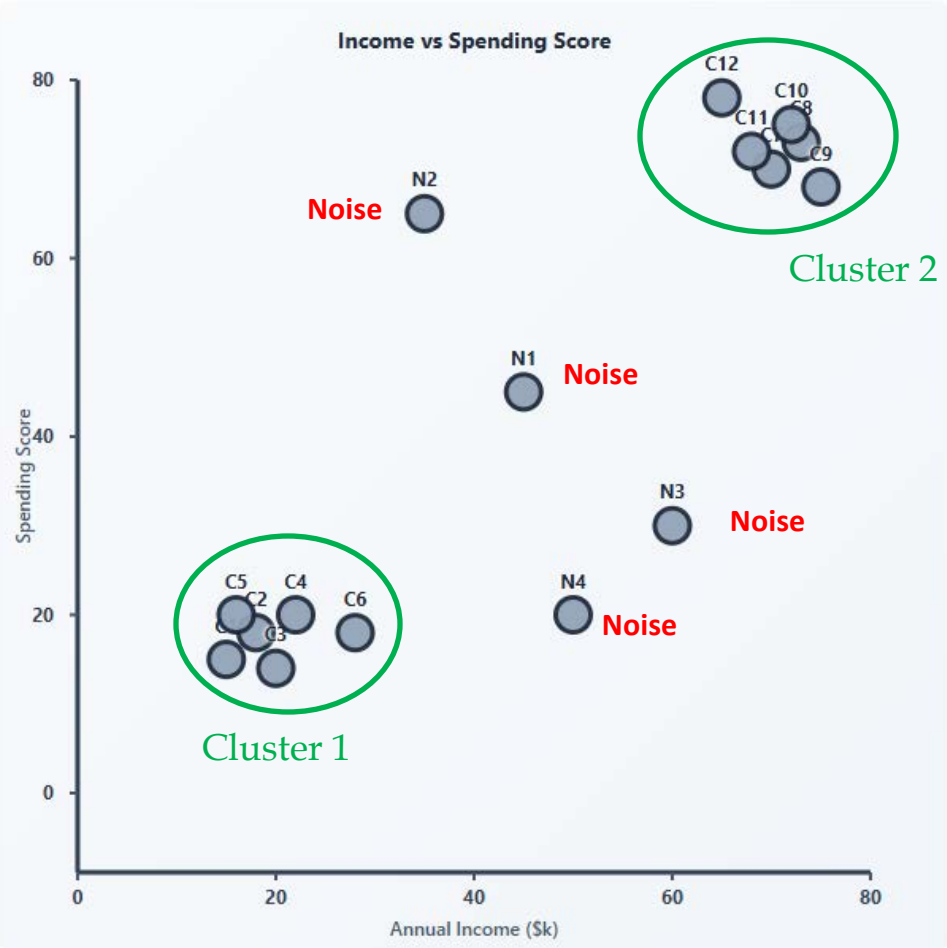
- **DBSCAN: Density-Based Spatial Clustering of Applications with Noise**
  - It is a density-based clustering algorithm that groups points into clusters based on local density, identifies noise points, and can discover clusters of arbitrary shape without specifying the number of clusters.

- Key parameters
  - ε (neighborhood radius) and
  - MinPts (minimum points to form a cluster).

- **Goal:**
  - Group together points that are **densely packed**, and mark points in **sparse regions as noise**.

What is DBSCAN?

- DBSCAN forms clusters by looking for areas where **many points are close together**.
- Points in sparse regions are treated as **noise**.
- Unlike K-Means, **we do NOT need to specify the number of clusters** in advance here



Key Concepts in DBSCAN

- **ε (Epsilon)**
  - The **maximum distance** between two points to be considered neighbors.
  - Controls the **radius** of a neighborhood.

- **MinPts (Minimum Points)**
  - The **minimum number of points** required to form a dense region (cluster).

Types of Points in DBSCAN

- **Core Point**
  - Has at least **MinPts neighbors within ε**.
- **Border Point**
  - Has fewer than MinPts neighbors but lies near a **core point**.
- **Noise Point (Outlier)**
  - Neither a core nor a border point.
  - Marked as **−1** in DBSCAN output.

How DBSCAN Works (Simple Steps)

- Pick an unvisited point.
- Check how many neighbors it has within ε.
- If neighbors ≥ MinPts → form a cluster.
- Expand the cluster by adding density-connected points.
- Repeat until all points are processed.

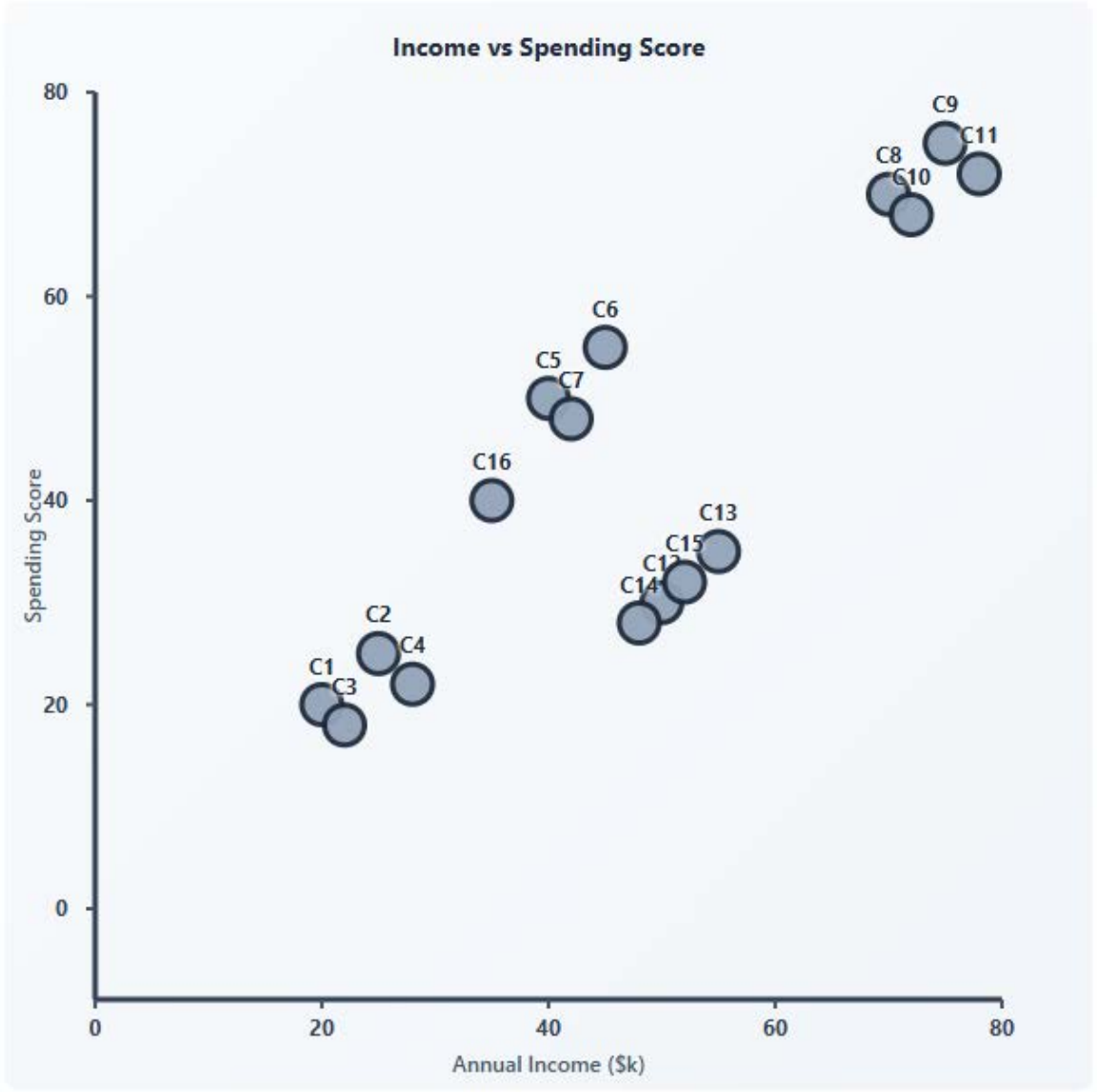
Example (Intuitive)

- Imagine people standing in a field:
  - People standing **close together** → form a cluster
  - People standing **alone** → considered noise

- DBSCAN naturally finds such groups.

Example – Clustering of Customers

Customer	Income ($k)	Spending Score
C1	15	39
C2	16	81
C3	17	6
C4	18	77
C5	19	40
C6	70	76
C7	71	94
C8	72	35
C9	14	50
C10	20	85
C11	16	12
C12	15	70
C13	68	88
C14	73	45
C15	69	65
C16	74	80



Example – Clustering of Customers

Initial Data
16 customers - DBSCAN will find clusters based on density

Parameters Set
ε (epsilon) = 12, MinPts = 3 - defines density threshold



Example – Clustering of Customers

Processing C1
Found 4 neighbors
within ε - CORE POINT!
Start Cluster 1 (Red)



Example – Clustering of Customers

Expanding Cluster 1
Adding neighbors C2, C3, C4 to Cluster 1



Example – Clustering of Customers

Processing C5

Found 3 neighbors - CORE POINT! Start Cluster 2 (Blue)



Example – Clustering of Customers

Expanding Cluster 2
Adding C6, C7, C16 to Cluster 2



Example – Clustering of Customers

Processing C8
Found 4 neighbors - CORE POINT! Start Cluster 3 (Green)



Example – Clustering of Customers

Expanding Cluster 3
Adding C9, C10, C11 to Cluster 3



Example – Clustering of Customers

Processing C12
Found 4 neighbors - CORE POINT! Start Cluster 4 (Orange)



Example – Clustering of Customers

Expanding Cluster 4
Adding C13, C14, C15 to Cluster 4



Example – Clustering of Customers

Final Result
4 clusters found!
DBSCAN automatically determined the number



Advantages and Disadvantages of DBSCAN

- **Advantages**
  - No need to choose number of clusters
  - Finds **arbitrary-shaped clusters**
  - Detects **outliers automatically**
- **Disadvantages**
  - Sensitive to choice of ε
  - Struggles with **varying densities**
  - Not ideal for very high-dimensional data

Thank You